Algebraic Geometry Notes

Alex Corbett

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1 Affine Algebraic Varieties

Definition 1.1. Let K be a field. Then an Affine Algrbraic Variety is a set of the form

$$\mathcal{V}(I) = \{(a_1, a_2, \ldots, a_n) \in K^n \mid f(a_1, a_2, \ldots, a_n) = 0, \forall f \in I\},$$

where I is an ideal $I \lhd K[x_1, x_2, \ldots, x_n]$.

Proposition 1.2. Let I_1, I_2 be ideals of $K[x_1, \ldots, x_n]$ be such that $I_1 \subseteq I_2$, then $\mathcal{V}(I_1) \supseteq \mathcal{V}(I_2)$.

Example 1.3.

(i) If $I = \{0\}$, then $\mathcal{V}(I) = K^n$.

(ii) If $I = K[x_1, x_2, \ldots, x_n]$, then $\mathcal{V}(I) = \{\emptyset\}$.

(iii) If $I = \langle x^2 + y^2 - 1\rangle \lhd \mathbb{R}[x, y]$, then $\mathcal{V}(I)$ is the circle of radius 1 centred at the origin defined by the equation $x^2 + y^2 = 1$ in $\mathbb{R}^2$.

(iv) Let $f_1, f_2, \ldots, f_r$ be polynomials of degree 1 in $x_1, x_2, \ldots, x_n$, and let $I = \langle f_1, f_2, \ldots, f_r\rangle$ be the ideal generated byu them. Then $\mathcal{V}(I)$ is the set of solutions of the system of linear equations $f_1 = f_2 = \cdots = f_r = 0$. Such a variety is called an affine subspace of K^n. This, in particular, includes sets containing a single point. If $P = (a_1, a_2, \ldots, a_n) \in K^n$, then the ideal $I = \langle x_1 - a_1, x_2 - a_2, \ldots, x_n - a_n\rangle$ defines the variety $\mathcal{V}(I) = \{P\}$.

(v) Let $I = \langle y^2 - x^3\rangle \lhd \mathbb{R}[x, y]$, then $\mathcal{V}(I)$ is the cuspoidal cubic curve defined by $y^2 = x^3$, also know as Neile's semicubal parabol, then $\mathcal{V}(I)$ is the cuspoidal cubic curve defined by $y^2 = x^3$, also know as Neile's semicubal parabol, then $\mathcal{V}(I)$ is the cuspoidal cubic curve defined by $y^2 = x^3$, also know as Neile's semicubal parabola.

(vi) Let $I = \langle y^2 - x^3 - x^2\rangle\mathbb{R}[x, y]$, then $\mathcal{V}(I)$ is the nodal cubic curve defined by the equation $y^2 = x^3 + x^2$.

(vii) If $I = \langle x, y\rangle \lhd [x, y]$, then $\mathcal{V}(I) = \{(0, 0)\} \subset \mathbb{R}^2$.

(viii) If $I = \langle x^2, y^2\rangle \lhd [x, y]$, then $\mathcal{V}(I) = \{(0, 0)\} \subset \mathbb{R}^2$.

(ix) If $I = \left\langle x^2 + y^2 + z^2 - 1, x^2 - x + y^2\right\rangle \lhd \mathbb{R}[x, y, z]$, then $\mathcal{V}(I) \subset \mathbb{R}^3$ is a figure of eight curve from an intersection of the cylinder, $y^2 = x^2 - x$, and the sphere of radius 1 at the origin.

(x) The special linear group $SL(n\mathbb{R})$ is also an affine algebraic variety. The set of all $n \times n$ real matrices is $\mathbb{R}^{n^2}$, the determinant is a polynomial in the entries of the matrix, and therefore $SL(n, \mathbb{R}) = \mathcal{V}(\langle \det(A) - 1\rangle)$.

Proposition 1.4. Let K be a field. Let $f_1, f_2, \dots, f_r \in K[x_1, x_2, \dots, x_n]$ and let $I = \langle f_1, f_2, \dots, f_r \rangle$. Then

$$\mathcal{V}(I) = \{(a_1, a_2, \dots, a_n) \in K^n \mid f_i(a_1, a_2, \dots, a_n) = 0, \forall 1 \leq i \leq r\}.$$

Proof. We know $f_i \in I$ for each i, so if $(a_1, \dots, a_n) \in \mathcal{V}(I)$, then $f_i(a_1, \dots, a_n) = 0$ for every i. Conversely, assume that $f_i(a_1, \dots, a_n) = 0$ for every i. Any $f \in I$ can be written as a $K[x_1, \dots, x_n]$-linear combination of the f_i, so

$$f(a_1, \dots, a_n) = \sum_{i=1}^{r} f_i(a_1, \dots, a_n) g_i(a_1, \dots, a_n) = 0,$$

so $(a_1, \dots, a_n)$. $\square$

Definition 1.5. A ring R is called Noetherian if all its ideals can be generated by finitely many elements.

Corollary 1.6 (Hilbert Basis Theorem). If K is a field, then the polynomial ring $K[x_1, \dots, x_n]$ is Noetherian for any $n \geq 0$.

Example 1.7.

(i) If R is Noetherian and $I \lhd R$, then R/I is also Noetherian.

(ii) $\mathbb{Z}$, the ring of integers, is Noetherian because it is a PID, so every ideal is generated by one element.

(iii) $K[x_1, x_2, \dots]$, the polynomial rinf over the field K in finitely many variables is an example of a non-Noetherian ring, because the ideal $\langle x_1, x_2, \dots \rangle$ is not finitely generated.

Proposition 1.8.

(i) Let $V_1 = \mathcal{V}(I_1), \dots, V_k = \mathcal{V}(I_k)$ be affine algebraic varieties in K^n. Then

$$V_1 \cup \dots \cup V_k = \mathcal{V}(I_1 \cap \dots \cap I_k) = \mathcal{V}(I_1 I_2 \cdots I_k)$$

is also an affine algebraic variety.

(ii) Let $V_\alpha = \mathcal{V}(I_\alpha), \alpha \in A$ be affine algebraic varieties in K^n. Then

$$\bigcap_{\alpha \in A} V_\alpha = \mathcal{V}\left(\sum_{\alpha \in A} I_\alpha\right)$$

is also an affine algebraic variety.

(iii) Let $V_1 = \mathcal{V}(I_1) \subseteq K^m, V_2 = \mathcal{V}(I_2) \subseteq K^n$ be affine algebraic varieties. Then $V_1 \times V_2 \subseteq K^m \times K^n \cong K^{m+n}$ is also an affine algebraic variety.

Proof.

(i) We first assume that $k = 2$. Let $P \in V_1 \cup V_2$ and let $f \in I_1 \cap I_2$. If $P \in V_1$, then $f(P) = 0$ because $f \in I_1$, and similarly if $P \in V_2$ then $f(P) = 0$ because $f \in I_2$. Therefore in either case we have $f(P) = 0$, so $P \in \mathcal{V}(1_1 \cap I_2)$, so $V_1 \cup V_2 \subseteq \mathcal{V}(I_1 \cap I_2)$.
We also know $I_1 I_2 \subseteq I_1 \cap I_2$, so we also have $\mathcal{V}(I_1 \cap I_2) \subseteq \mathcal{V}(I_1 I_2)$.
Let now $P \in K^n \setminus (V_1 \cup V_2)$. Then $P \notin V_1$, so there exists $f_1 \in I_1$ such that $f_1(P) \neq 0$. Similarly, $P \neq V_2$ so there exists $f_2 \in I_2$ such that $f_2(P) \neq 0$. Then $f_1 f_2 \in I_1 I_2$ and $(f_1 f_2)(P) = f_1(P) f_2(P) \neq 0$, therefore $P \notin \mathcal{V}(I_1 I_2)$. This implies $\mathcal{V}(I_1 I_2) \subseteq V_1 \cup V_2$. Hence we obtain

$$V_1 \cup V_2 = \mathcal{V}(I_1 \cap I_2) = \mathcal{V}(I_1 I_2).$$

For $k > 2$ we use induction on k.

$\square$